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Differential Privacy

A story about people, security measures, mathematical chaos, and what black magic can tell us about them.

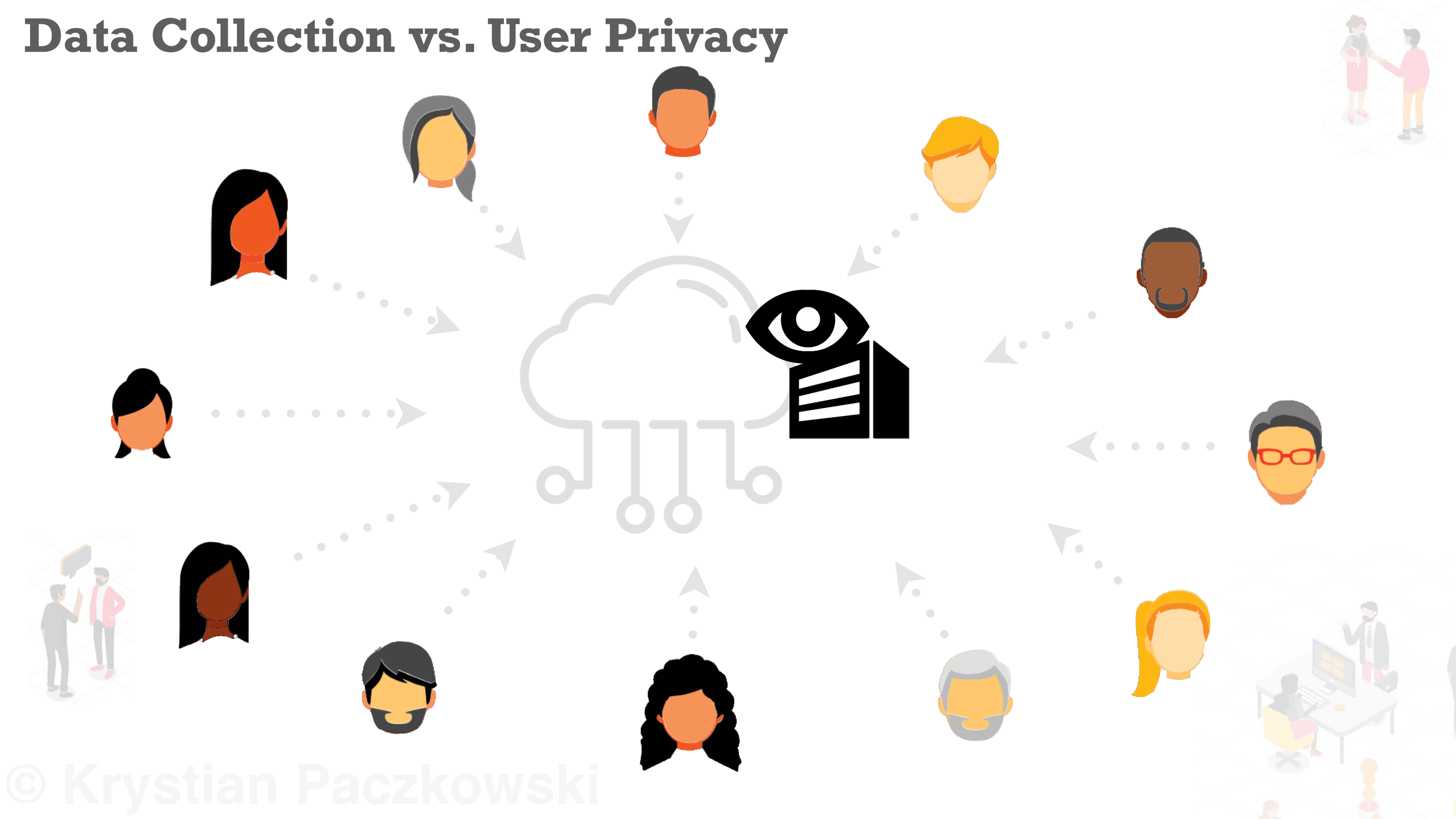
Krystian Paczkowski

March 27, 2020



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Data Collection vs. User Privacy



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Data Anonymization



ZIP CODE



GENDER



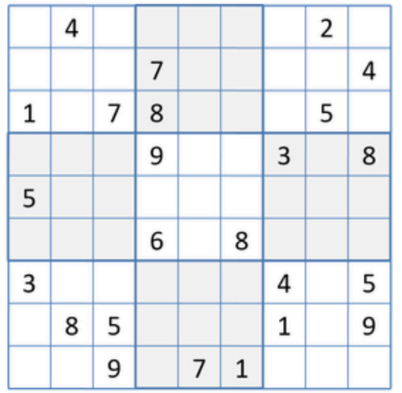
BIRTHDAY

PROBLEMS

- We do not know **WHAT** kind of questions will be asked?
- We do not know **WHO** will be asking?
- We need to assure there is **no correlation!**
- Is it really **anonymous**?

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RISKS
Privacy Breach



★ **Linkage attack** (correlation to public databases):
- Latanya Sweeney's de-anonymizing attack connecting voter record to anonymized medical data
- Netflix challenge dataset (combined with IMDb)

★ **Differencing attack**

We can reach information through:

```
SELECT AVG(salary) WHERE name = 'John Doe'
```

If query limited to set of T=2

```
SELECT SUM(salary);
SELECT SUM(salary) WHERE name != 'John Doe';
```

Or knowing John Doe is 40 years old:

```
SELECT SUM(salary) WHERE position = 'WR';
SELECT SUM(salary) WHERE position = 'WR' AND age != 40;
```

★ **Reconstruction attack**
- CENSUS: re-identification of 308,745,548 (2010) successful linkage to commercial data

K-anonymity technique - based on information of K other individuals a person can be re-identified. Not good with high-dimensional data; shows that using just 4 locations of mobile phone timestamp-location can be as high as 95%. **https://arxiv.org/pdf/1709.05772.pdf**

TRADITIONAL
Anonymization is NOT a solution (re-identification attacks)

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RISKS
Security Breach

Security:

- **INTERNAL:** user needs low level data access;
  by changing permissions this could result in *insider attack*

- **EXTERNAL:** Organization shares data with peers, contractors, government:
  inference of private information

Databases can be:

- **INTERACTIVE**: a process (privacy manager) sits in front of the database that
  can analyze a query before applying the privacy mechanism

  - **NON-INTERACTIVE** (when we want to release it to somebody):
    - data has been anonymized (de-identified) by removing names, zip codes etc.
    - perturbed
    - sampled



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SOLUTION



Differential Privacy

Neutralizes effect of aforementioned attacks

Laplace distribution
aims to spread data over larger range.

Approach assumes the results will be the same whether we participate or not

Formal Privacy Guarantee

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Differential Privacy

Cynthia Dwork (2006)
(Microsoft; patented)



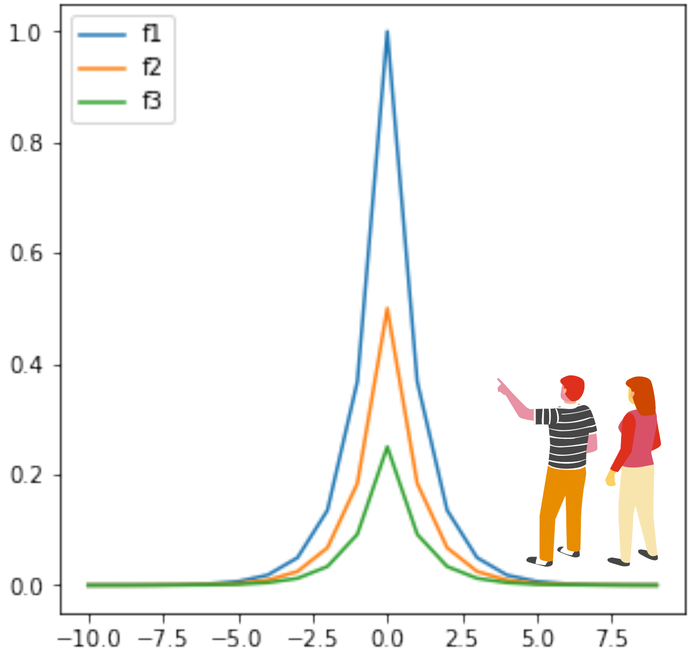
Goal: *Allow users to learn about the population as the WHOLE at the same time protecting information about the individual representation that could not be learned without dataset access. At the same time, risk of privacy shall not increase as a result of being a member of a database.*

- ★ The presented method does not use cryptography (do not confuse with encryption), secure function computation, etc. Such approaches are generally still vulnerable;
- ★ Involves trade-offs between privacy and accuracy; to maintain the privacy noise is introduced maintaining "close-enough" correlation to the real answer;
- ★ Utility and privacy budgets

  The more information you query, the more noise has to be injected to minimize the privacy leakage.
- ★ Protects **outliers**

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Math behind it



- The Laplace distribution is a continuous probability distribution also called the double exponential distribution;
- The difference between two independent identically distributed variables is governed by a Laplace distribution.

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Differential Privacy

Adversary having access to (anonymized) database has no advantage over adversary that knows nothing.

$$Pr[\kappa(D_1) \in S] \leq e^{\epsilon} \times Pr[\kappa(D_2) \in S]$$

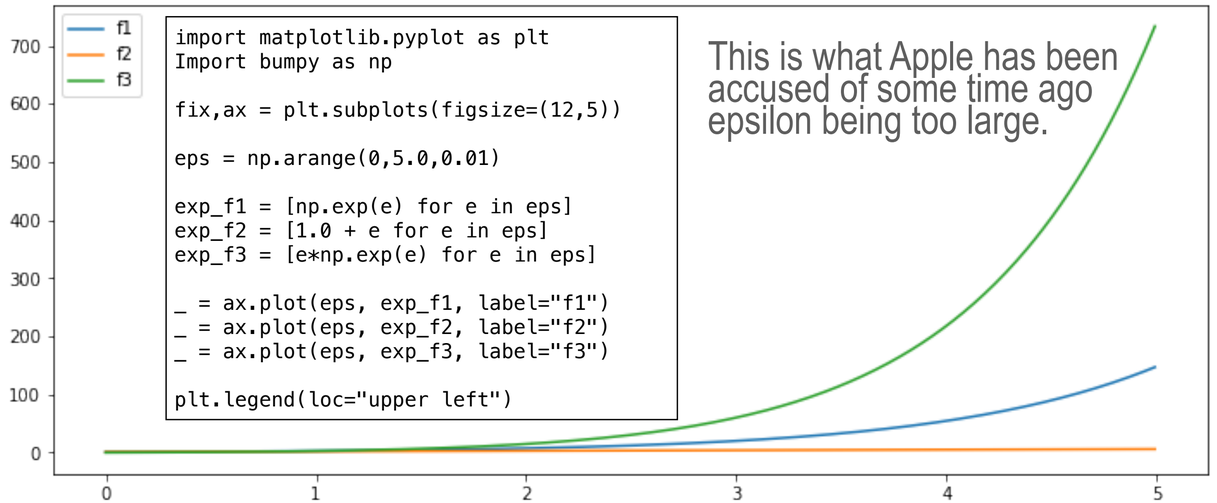
Scale: privacy budget defined by sensitivity

(2006; Cynthia Dwork)

κ - privacy mechanism function

In human digestible format: the probability of particular query on D1 is the same (within a small factor) of the probability of getting the same result on D2.

Easy to see probability of D1 and D2




```
import matplotlib.pyplot as plt
Import bumpy as np

fix,ax = plt.subplots(figsize=(12,5))

eps = np.arange(0,5.0,0.01)

exp_f1 = [np.exp(e) for e in eps]
exp_f2 = [1.0 + e for e in eps]
exp_f3 = [e*np.exp(e) for e in eps]

_ = ax.plot(eps, exp_f1, label="f1")
_ = ax.plot(eps, exp_f2, label="f2")
_ = ax.plot(eps, exp_f3, label="f3")

plt.legend(loc="upper left")
```

This is what Apple has been accused of some time ago epsilon being too large.



Desired output: Hard to distinguish results from one database and the other.
Task: Same query on both databases
Process: Run privacy mechanism on both databases
Yield: Answer: our hope is it would be difficult to distinguish both results, such as one result is different from the other, but still represents what we need

Right ϵ is the key!

- small epsilon yields better privacy
- ratio of datasets difference and epsilon defines data leakage
- $\epsilon \times e^{\epsilon} \approx (1 + \epsilon)$
- with small epsilon we might add noise hence affect accuracy!

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Differential Privacy
SENSITIVITY

Differential Privacy is accomplished by an addition of a **random noise**, where the magnitude of the noise is a function of the largest changed within a single sample (participant of the population) that can occur upon the output of the query.

Noisy solution is a delicate solution, if the noise is symmetric about the origin and the same question is asked many times the responses may be averaged, canceling out the noise!

The sensitivity of privacy is: κ - privacy mechanism function ϵ - privacy budget

$$\Delta\kappa = max_{D_1,D_2} = ||\kappa(D_1) - \kappa(D_2)||_1$$

If the difference only in one term/row the response histogram will only change in one bin (k/ϵ = $1/\epsilon$) Hence sensitivity = 1

When k = 1 the sensitivity is the maximum difference in the values that privacy mechanism may take on a pair of databases.

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Differential Privacy
NOISE aka Laplacian

For every query our function adds noise with distribution equal to $Lap(\Delta\kappa/\epsilon)$

$$f(X) + (Lap(\Delta\kappa/\epsilon)^k$$

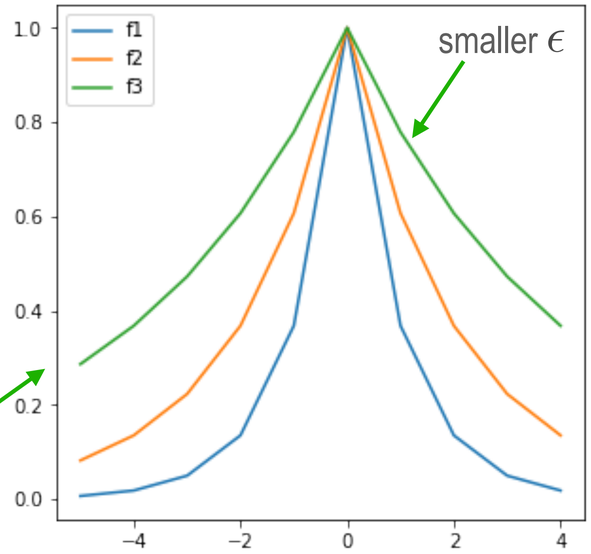
Notice: decreasing ϵ flattens the Laplacian curve, hence yielding larger expected noise

Question:

How much Laplacian noise do we add to achieve differential privacy:

$$\epsilon = \Delta\kappa/Var(X)$$

Laplacian Noise:




```
import matplotlib.pyplot as plt
import numpy as np

fix,ax = plt.subplots(figsize=(5,5))

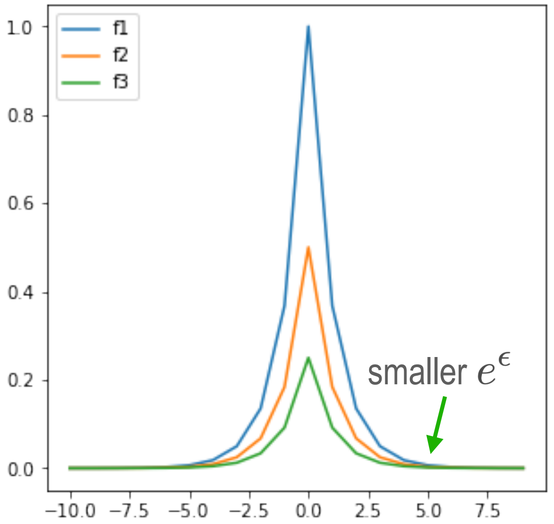
eps = np.arange(-5.0,5.0,1.0)

exp_f1 = [np.exp(-np.abs(e)) for e in eps]
exp_f2 = [np.exp(-np.abs(e/2)) for e in eps]
exp_f3 = [np.exp(-np.abs(e)/4) for e in eps]

_ = plt.plot(eps, exp_f1, label="f1")
_ = plt.plot(eps, exp_f2, label="f2")
_ = plt.plot(eps, exp_f3, label="f3")

plt.legend(loc="upper left")
```

Mechanism picks random value from Laplacian distribution, each time asking the same query we will get different value. It can get close or far from the real value or far (i.e. within +/- 10 off true value)




```
import matplotlib.pyplot as plt
import numpy as np

fix,ax = plt.subplots(figsize=(5,5))

eps = np.arange(-10.0,10.0,1.0)

exp_f1 = [np.exp(-np.abs(e)) for e in eps]
exp_f2 = [np.exp(-np.abs(e))/2 for e in eps]
exp_f3 = [np.exp(-np.abs(e))/4 for e in eps]

_ = plt.plot(eps, exp_f1, label="f1")
_ = plt.plot(eps, exp_f2, label="f2")
_ = plt.plot(eps, exp_f3, label="f3")

plt.legend(loc="upper left")
```

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Differential Privacy

ϵ - privacy budget

Laplacian Noise:

sensitivity

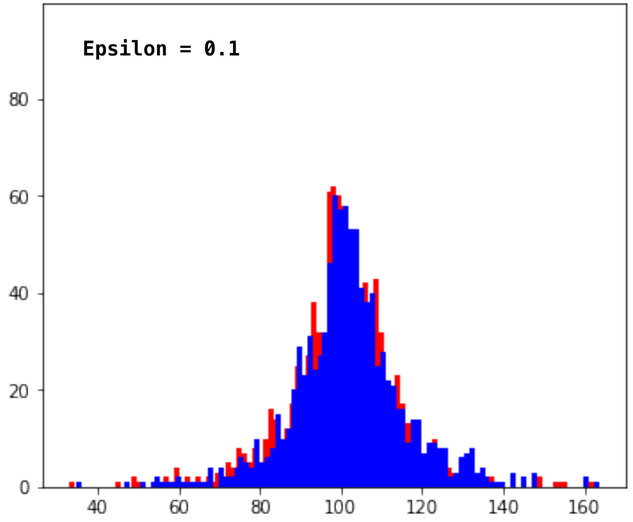
```
def plot_laplacian(ax, eps):
    lap1 = np.random.laplace(100.0, 1.0 / eps, 1000)
    _ = ax.hist(lap1, bins=100, color="red")
    lap2 = np.random.laplace(101.0, 1.0 / eps, 1000)
    _ = ax.hist(lap2, bins=100, color="blue")

f, (ax1, ax2, ax3) = plt.subplots(1, 3, sharey = True, figsize=(20,5))

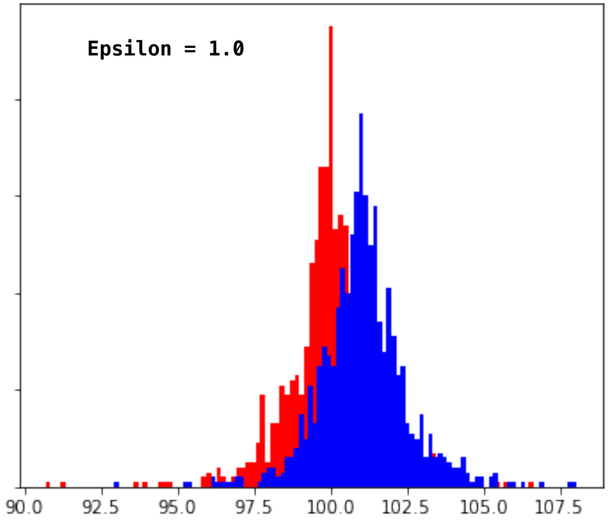
ax = [ax1,ax2,ax3]
eps = [0.1,1.0,10.0]

for i in range(len(eps)):
    plot_laplacian(ax[i],eps[i])
```

Note: Notice each Laplacian calculation always yields different answers

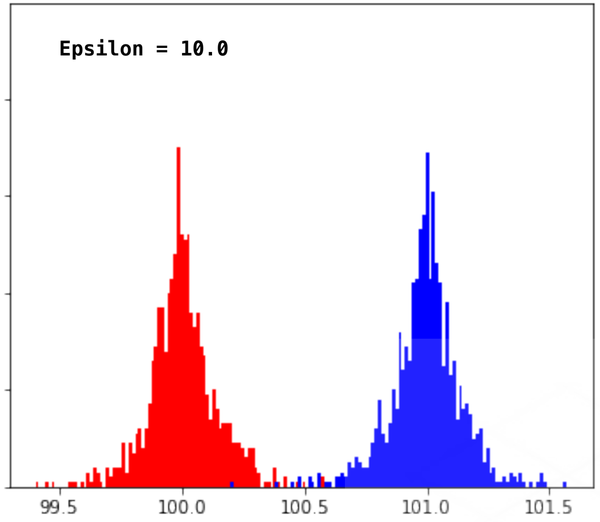


All the points overlap, so generally, by selecting two different points in each dataset, we cannot tell the difference from which distribution the point came from.



If we pick just a single point from both, most likely we cannot tell from which population the point query came from.

Nonetheless, if we can make enough queries in those databases we can see there is a difference, such as mean in those two things.



We can very easily tell from which distribution the point came from.

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Differential Privacy
RISKS

We need to add more noise, when:

- ★ number of queries is high,
- ★ multiple people work on the same data base.

We add more information on the queries, but we also need to limit amount of queries as not to exceed the ***privacy budget***.

Each query yields more information and increases the privacy risk by reducing the budget.

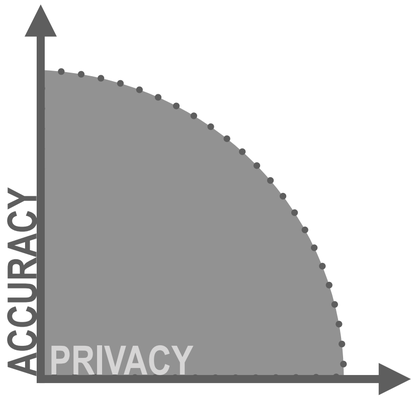
$$\epsilon_{TOTAL} = \epsilon_1 + \epsilon_2 + ... + \epsilon_n$$

Hence, differential privacy limits user to allowable privacy budget that limits number of queries $\Longrightarrow$ budget for number of queries plus a limit of people to collaborate with each other.



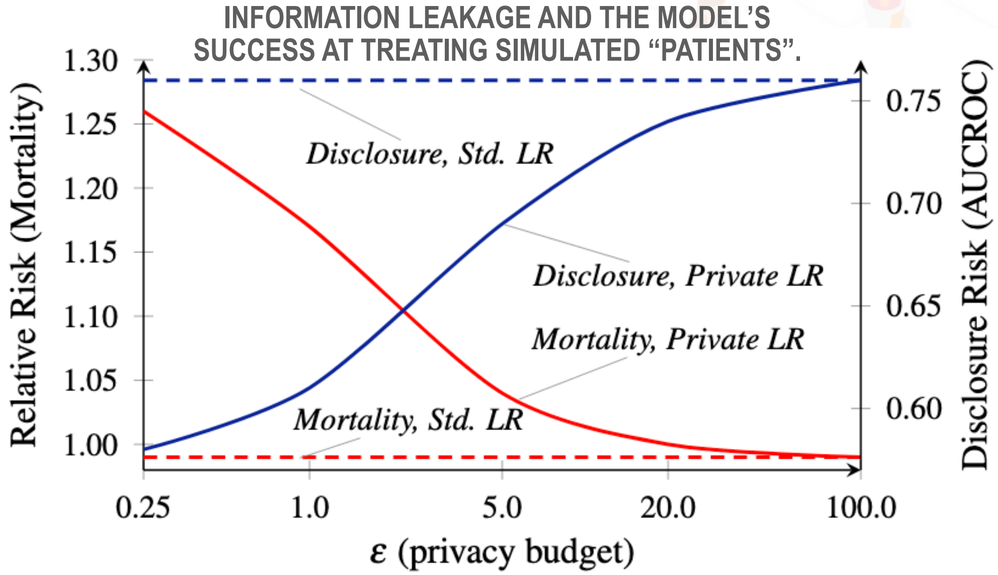
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Differential Privacy
RISKS: TRADE-OFFs



- The authors linked medicine dosage to specific genetic markers with ML techniques to develop a dosing model; DP applied at various privacy budgets while training the model.
- The results showed that the model's accuracy depends a lot on the privacy budget on which it was trained.

- If the budget is set too high, the database leaks a great deal of sensitive patient information, but the resulting model makes dosing decisions that are about as safe as standard clinical practice.
- When the budget was reduced to a level that achieved meaningful privacy, the "noise-ridden" model had a tendency to **kill** its "patients".



Frederikson et al. from 2014:
https://www.usenix.org/system/files/conference/usenixsecurity14/sec14-paper-fredrikson-privacy.pdf

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Differential Privacy
RISKS

ACCURACY is affected by:

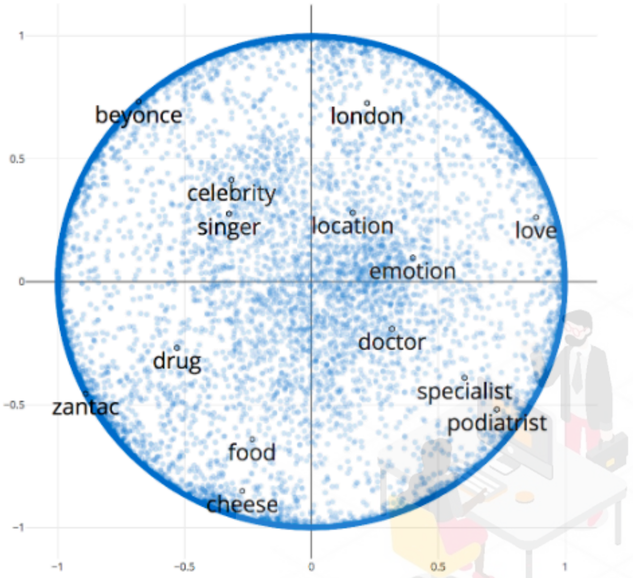
- ★ % of population subjected to calculation
- ★ diversity of the population
- ★ type of calculation
- ★ number of performed calculations
- ★ privacy budget $\begin{cases} \epsilon \to 0 & \text{complete privacy} \\ \epsilon \to \infty & \text{fully identifiable population} \end{cases}$

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Differential Privacy
NLP - RISKS

- machine learning models are very good at memorizing training data (individual points) that can be reverse engineered.

- In NLP, through foundations in generative models, the risk of prediction of the training data from the outputs is quite high. The data used to train *language models* can be reverse engineered giving users of the models access to part of the training data.



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Differential Privacy

NLP - Bag of Words

February 2019

- ★ "Obfuscate" texts by removing stylistic clues which can identify authorship, while preserving the content of the text.
- ★ Text documents represented as "bags-of-words". Sufficient for *topic identification* and *authorship attribution* (of the original documents).

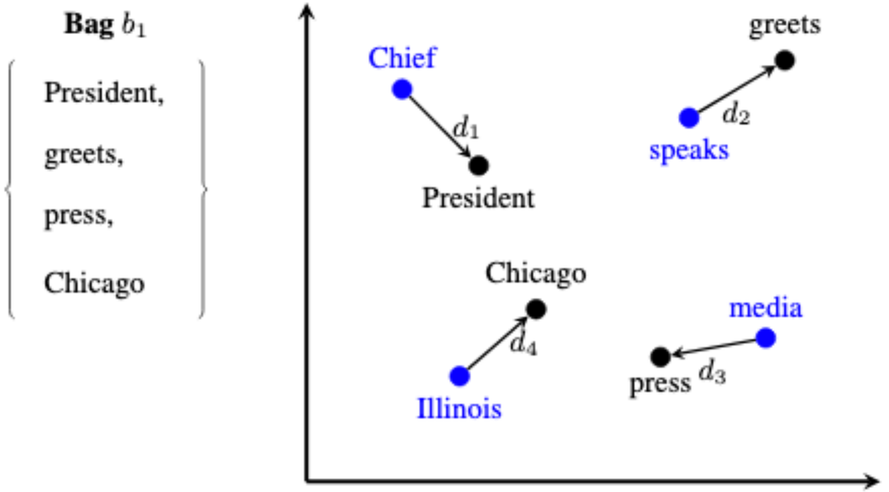


Fig. 1: Earth Mover's metric between sample documents.

- ★ Mechanism satisfies privacy with respect to a metric for *semantic similarity* with the obfuscation of stylistic clues.
- ★ Demonstrated implementation on a "fan fiction" dataset.
- ★ **It is possible to disguise writing style and preserve document information for accurate content classification.**

https://link.springer.com/chapter/10.1007/978-3-030-17138-4_6
https://arxiv.org/pdf/1811.10256.pdf

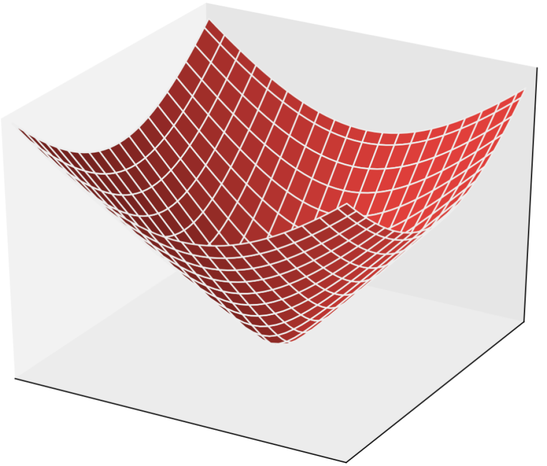
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Differential Privacy

NLP

AMAZON Research
November, 2019

- Word representations in *hyperbolic space* as means of preserving privacy
- Re-phrasing the customer-supplied text and basing analysis on the new phrasing, rather than on the customers' own language.
- Probability distribution in *hyperbolic space* and describe a way to sample from it in high-dimensions.



  - Privacy is provided by perturbation of word vector representations in high dimensional *hyperbolic space.*
    - Authors observe - compared to the Euclidean space - there is *x20* greater guarantee on expected privacy against comparable worst case scenario.

https://www.amazon.science/publications/leveraging-hierarchical-representations-for-preserving-privacy-and-utility-in-text

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Differential Privacy

NLP

- This noise resulted in a loss of accuracy, but the expectation is that as the size of the data set grows, the trade-off between usefulness and privacy will become more manageable.

https://www.amazon.science/publications/leveraging-hierarchical-representations-for-preserving-privacy-and-utility-in-text

AMAZON Research
January 23, 2020

- Authors analyzed the privacy implications of different choices of epsilon in the context of natural language processing, where word embeddings often depend on the frequency with which words co-occur.
- Adding noise to a word embedding produces a new point; its location is not a valid word embedding; search is performed to find the nearest valid embedding.

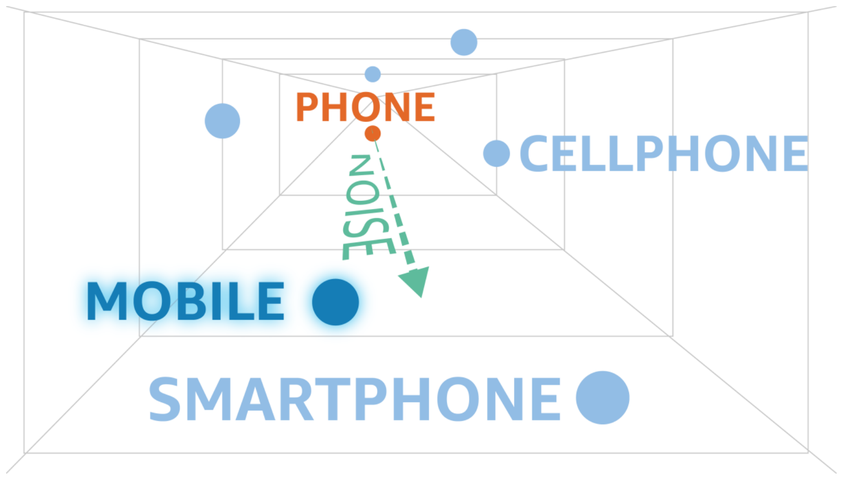
  - Authors attempt to predict the likelihood of a word being overwritten by noise + number of semantically related words
  - Word positions along the curve indicate which are more specific terms and which are more general.

https://www.amazon.science/blog/preserving-privacy-in-analyses-of-textual-data

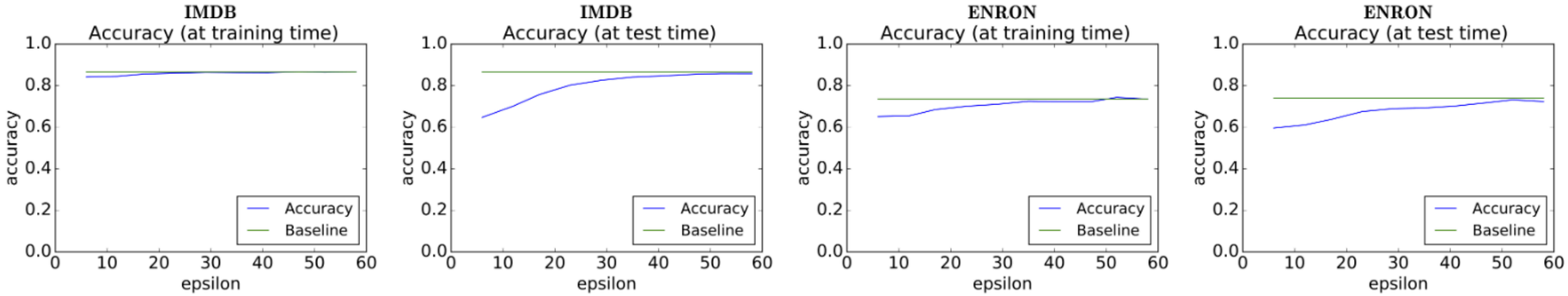
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Differential Privacy
NLP

AMAZON Research
January 23, 2020



		w = encryption		w = hockey		w = spacecraft	
ε	Avg. N_w	GloVe	fastText	GloVe	fastText	GloVe	fastText
⟵ increasing ε, better semantics	50	freebsd multibody 56-bit public-key	ncurses vpns tcp isdn	stadiumarena futsal broomball baseballer	dampener popel decathletes newsweek	telemeter deorbit airbender aerojet	geospace powerup skylab unmanned
	100	ciphertexts truecrypt demodulator rootkit	plaintext diffie-hellman multiplexers cryptography	interleague usrowing football lacrosse	basketball lacrosse curlers usphl	laser apollo-soyuz agena phaser	voyager cassini-huygens adrastea intercosmos
	200	harbormaster unencrypted cryptographically authentication	cryptographic ssl/tls authentication cryptography	players ohl goaltender defenceman	goaltender ephl speedskating eishockey	launch shuttlecraft spaceborne interplanetary	orbited tatooine flyby spaceborne
	300	decryption encrypt encrypted encryption	encrypt unencrypted encryptions encrypted	nhl hockeydb hockeyroos hockey	hockeygoalies hockeyroos hockeyettan hockey	spaceplane spacewalk spaceflights spacecraft	spaceship spaceflights satellites spacecraft



https://assets.amazon.science/4a/27/e1438fc948a690ff3c94ff19a7a1/privacy-and-utility-preserving-textual-analysis-via-calibrated-multivariate-perturbations.pdf

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Differential Privacy
OTHER APPROACHES

Google Research
October 2016

- Conservative noise to the parameters, where the noise is selected according to the worst-case analysis, can destroy the utility of the learned model.

- Authors control the influence on the training data during the training process, specifically in the SGD computation.

  - At each step of the SGD, during gradient calculation an average is calculated and noise is added to protect privacy.

  - At the end, in addition to output model, the privacy loss of the mechanism is also calculated.

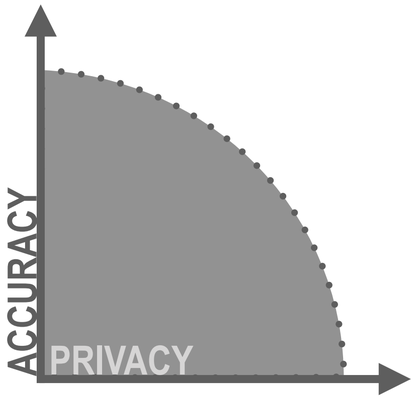
Ian Goodfellow et al.: https://arxiv.org/pdf/1607.00133.pdf

CODE: github.com/tensorflow/models: DPSGD_Optimizer()
https://github.com/tensorflow/privacy

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Differential Privacy

TAKEAWAYS



- ★ Attacker's access to information is the same as if she/he did not have access to data or if given entry was absent from the dataset.
- ★ Differential Privacy (DP) can be accomplished by *adding random noise*; the noise protects individual data introducing small variability
- ★ One of the algorithms to handle it is via *Laplace distribution*

- ★ We can introduce precise *amount of noise* and *risk* into the data that relies on *sensitivity* measure.
- ★ Allows us to estimate precise amount of re-identification risk via *privacy budget*

- ★ *Privacy budget* is not a weakness; if yes, then data should never be released

  $\epsilon \to 0$ complete privacy $\epsilon \to \infty$ fully identifiable population

- ★ Differentially private algorithms are still an active field of research.

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Differential Privacy

When will we use it?
Why not so far?

Requirments:

- ★ Need of large data set; will not work on small dataset

Challenges:

- ★ (deployment: needs to work with different databases)
- ★ no good usability for non-experts
- ★ board support for analytical queries
- ★ easy integration with existing data environments

Things to consider:

- ★ Privacy budget
- ★ Utility budget

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Differential Privacy

Possible use cases within

could:

- ★ We can learn about users without compromising privacy;
- ★ Could be used on global studies on trends within PwC.
  (user metrics, how we utilize machines, computational resources, deployments, etc.)

Not to exhaust privacy budget:

- ★ inquiries limited to monthly or annual data inputs.
- ★ once accessed the queries could be used for multiple studies; hence, not exceeding original privacy budget.
- ★ At worse, privacy losses add up; we exhaust inquiries.

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Who is using it?

- ★ **Microsoft: PINQ (Privacy Integrated Queries)**
- ★ **Apple (mobile devices)**- Internally only
- ★ **Google (search)**- Internally only (Chrome)
- ★ **Facebook (clicks)**- Internally only
- ★ **UBER — Open Source Library CHORUS (SQL bases)**
- ★ **IBM (`sklearn` based)**
- ★ **CENSUS 2020**

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Open Source libs?

IBM

https://github.com/IBM/differential-privacy-library

Scikit-Learn:

```
from sklearn import datasets
from sklearn.model_selection import train_test_split

dataset = datasets.load_iris()
X_train, X_test, y_train, y_test = train_test_split(dataset.data, dataset.target, test_size=0.2)
```

```
import diffprivlib.models as models

clf = models.GaussianNB()
clf.fit(X_train, y_train)
clf.predict(X_test)
```




`Diffprivlib` is made of three (3) modules:

- **Mechanisms:** These are the building blocks of differential privacy, and are used in all models that implement differential privacy. Mechanisms have little or no default settings, and are intended for use by experts implementing their own models. They can, however, be used outside models for separate investigations, etc.
- **Models:** module includes machine learning models with differential privacy. Library currently has models for clustering, classification, regression, dimensionality reduction and pre-processing.
- **Tools:** comes with a number of generic tools for differentially private data analysis. This includes differentially private histograms, following the same format as Numpy's histogram function.

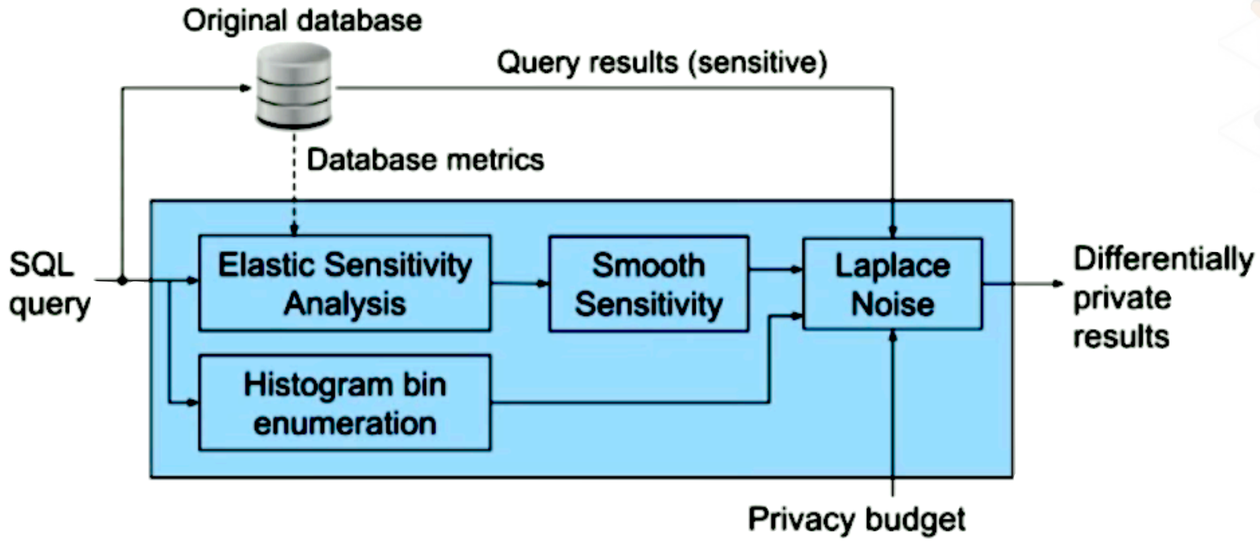
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Open Source libs?

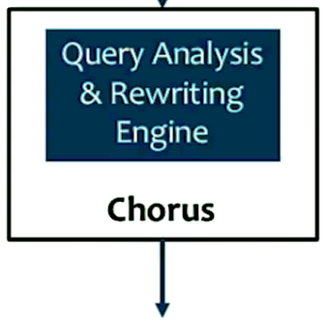
CHORUS (UBER)

Uber's blog post: https://medium.com/uber-security-privacy
CODE: https://github.com/uber/sql-differential-privacy

- ★ Tested on ~8mln entries
- ★ 18,774 SQL diff. queries




```
SELECT COUNT(*) FROM trips
```




```
SELECT COUNT(*) + 1/ε * SIGN(RAND())*LN(1-2*ABS(RAND()))
FROM trips
```

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Open Source libs?

Google

https://github.com/google/differential-privacy/

C++ library:

```
#include "differential_privacy/algorithms/count.h"

// Epsilon is a configurable parameter. A lower value means more privacy but
// less accuracy.
int64_t count(const vector<double>& values, double epsilon) {
  // Construct the Count object to run on double inputs.
  std::unique_pointer<differential_privacy::Count<double>> count =
     differential_privacy::Count<double>::Builder().SetEpsilon(epsilon)
                                                   .Build()
                                                   .ValueOrDie();


  // Compute the count and get the result.
  differential_privacy::Output result =
     count->Result(values.begin(), values.end());

  // GetValue can be used to extract the value from an Output protobuf. For
  // count, this is always an int64_t value.
  return differential_privacy::GetValue<int64_t>(result);
}
```


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Google

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PostgreSQL (11) Extension

In PostgreSQL, load the extension by calling:

```
CREATE EXTENSION anon_func;
```

Count

```
ANON_COUNT(column)
ANON_COUNT(column, epsilon)
```

Sum (`SUM`), Average (`AVE`), Variance (`VAR`), Standard Deviation (`STDDEV`)

```
ANON_SUM(column)
ANON_SUM(column, epsilon)
ANON_SUM_WITH_BOUNDS(column, lower, upper)
ANON_SUM_WITH_BOUNDS(column, lower, upper, epsilon)
```


Also read about RAPPOR project

In this table, each row represents one fruit eaten. If person 1 eats 2 apples, then there will be two rows in the table with column values (1, apple). Consider a simple query counting how many items of each fruit have been eaten.



```
SELECT fruit, COUNT(fruit)
FROM FruitEaten
GROUP BY fruit;
```

Suppose that instead of getting the regular count, we want the differentially private count with the privacy parameter ε=ln(3). The final product of the query rewrite would be

```
SELECT result.fruit, result.number_eaten
FROM (
  SELECT per_person.fruit,
    ANON_SUM(per_person.fruit_count, LN(3)/2) as number_eaten,
    ANON_COUNT(uid, LN(3)/2) as number_eaters
    FROM(
      SELECT * , ROW_NUMBER() OVER (
        PARTITION BY uid
        ORDER BY random()
      ) as row_num
      FROM (
        SELECT fruit, uid, COUNT(fruit) as fruit_count
        FROM FruitEaten
        GROUP BY fruit, uid
      ) as per_person_raw
    ) as per_person
  WHERE per_person.row_num <= 5
  GROUP BY per_person.fruit
) as result
WHERE result.number_eaters > 50;
```


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Open Source libs?

JAVA Sample:

```
import org.apache.commons.math3.distribution.LaplaceDistribution;

double laplaceMechanismCount(long realCountResult, double epsilon) {
    LaplaceDistribution ld = new LaplaceDistribution(0, 1 / epsilon);
    double noise = ld.sample();
    return realCountResult + noise;
}
```

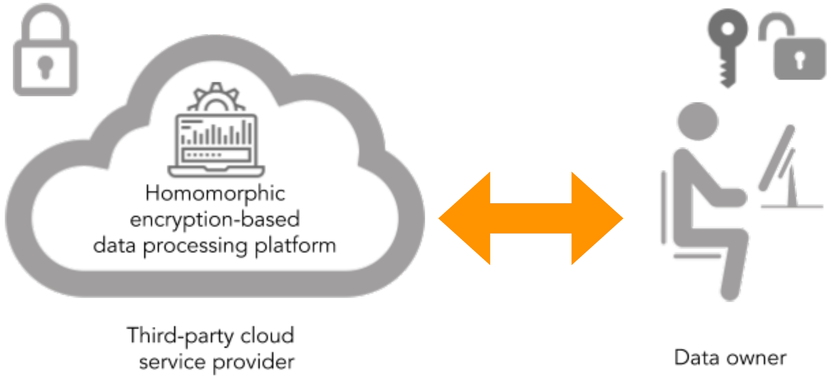

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Other options?

★ Federated Learning



★ Homomorphic Encryption



- ★ DARPA funded **PALISADE**
- ★ **SEAL** (Microsoft)
- ★ **HElib** (developed by cryptography icons)

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Differential Privacy



THANK YOU



Krystian Paczkowski

March 27, 2020



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